Delta Lake Incremental Data Processing Assignment

Student Information

Field	**Details**
Assignment	Delta Lake Incremental Data Processing
Technology	Apache Spark, Delta Lake, Databricks
Notebook	delta_scd_assignment.ipynb

Objective

The objective of this assignment was to perform incremental data processing using Delta Lake in Databricks. The assignment demonstrates how Delta Lake supports efficient data ingestion, data cleaning, and incremental updates using the MERGE operation.

Tools and Technologies

- Databricks
- Apache Spark (PySpark)
- Delta Lake
- CSV Dataset
- Unity Catalog / Delta Tables

Implementation Steps

Step 1 – Data Ingestion

- Loaded the customer dataset from a CSV file.
- Read the data into a Spark DataFrame.
- Created a managed Delta table named **customer_delta**.

Outcome

The raw customer data was successfully stored as a Delta table.

Step 2 – Data Cleaning

The following data quality operations were performed:

- Identified missing values.
- Replaced null values with appropriate default values.
- Removed duplicate records.
- Overwrote the Delta table with cleaned data.

Outcome

The dataset became cleaner and suitable for further processing.

Step 3 – Incremental Dataset Creation

To simulate incoming customer data:

- Existing customer records were modified.
- New customer records were generated.
- The records were combined into an incremental dataset.
- The dataset was saved as **customer_incremental.csv**.
- A second Delta table named **customer_incremental_delta** was created.

Outcome

A realistic incremental dataset was prepared for processing.

Step 4 – Delta MERGE Operation

The incremental Delta table was merged into the main Delta table.

The MERGE operation performed:

- Update existing customer records.
- Insert new customer records.

Outcome

The customer Delta table was successfully updated without creating duplicate records.

Step 5 – Validation

The following validations were performed:

- Verified total record count.
- Checked duplicate customer IDs.
- Verified updated customer records.
- Verified newly inserted customer records.

Outcome

The Delta table contained the expected updated and inserted records.

Step 6 – Final Output

The final customer Delta table was displayed after completing all processing steps.

The assignment successfully demonstrated an end-to-end incremental data processing pipeline using Delta Lake.

Key Delta Lake Features Used

- Delta Tables
- ACID Transactions
- MERGE Operation
- Data Cleaning
- Incremental Processing
- Managed Tables

Learning Outcomes

After completing this assignment, the following concepts were understood:

- Reading CSV files using Spark
- Creating Delta tables
- Data cleaning techniques
- Handling null values
- Removing duplicate records
- Creating incremental datasets
- Performing Delta MERGE operations
- Validating processed data

Conclusion

This assignment demonstrated how Delta Lake simplifies incremental data processing by allowing updates and inserts through the MERGE operation while maintaining data consistency. The implementation followed a practical ETL workflow, making it suitable for real-world data engineering scenarios.